

Share Your Knowledge

If you are an expert in your field of electronics, why not share your knowledge with the world?



Electronics For You invites electronics professionals and industry experts to write articles on their area of expertise and interest from different perspectives

Some of the topics we want you to write about are:

- Photonics: Optics & Optoelectronics
- Automation
- Power Electronics
- Components
- Telecom

So if you have an interesting topic or original article to share, please let us know at efyeditorial@efyindia.com

We would love to hear your good ideas too!

electronics
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Task & responsibilities of an instrumentation engineer

- Develop concept designs and feasibility studies
- Design or update specifications and designs of electrical systems and equipment
- Design ways to generate and manage electricity
- Devise ways to control processes using automation techniques and computers
- Examine safety aspects of the planned work
- Plan manufacture or use of equipment and appliances
- Oversee testing of new systems and correction of faults
- Supervise the installation and maintenance of systems and equipment
- Design new control systems, and develop, maintain and modify existing systems
- Work collaboratively with design engineers, operation engineers, purchasers and other internal staff
- Contact clients, suppliers, contractors and relevant authorities
- Project management within cost- and time-constrained environments
- Write computer software
- Develop new business proposals
- Liaise with clients, suppliers, contractors and relevant authorities, e.g., the Nuclear Decommissioning Authority
- Understand and ensure compliance with health and safety regulations and quality standards of the country in which work is undertaken

aerospace also require instrumentation engineers," says Kamat.

Instrumentation engineering comprises design and development of more efficient electrical machinery, control equipment and power systems for generation, transmission and distribution of electrical energy, and telecommunication. Instrumentation technicians perform operations like commissioning, installation, calibration, maintenance and repair of complex measurement and control systems used in a variety of process control operations. Those willing to make a career in this field should have a sound knowledge of measurement and control principles of pneumatic and electronic instruments theoretically as well as practically. Tasks and responsibilities of instrumentation engineers vary from company to company (please see the box).

Qualification required

A BE or B.Tech degree is desirable. However, a diploma course in instrumentation engineering after passing 10th standard will be sufficient for becoming a junior engineer. Even 10+2 passouts in science stream with physics, chemistry and mathematics as main subjects are hired. If research interests you, enroll in a Ph.D program

after completing your post-graduation. Besides basic sciences and regular engineering subjects, microwave communication, radar and television engineering, microelectronics, computers, and communication and control systems are taught at graduation level.

"Instrumentation engineers can be science graduates in physics, chemistry and geology, engineering diploma holders or graduates in electronics, instrumentation and electrical engineering. Mechanical and chemical engineers with exposure to electronics can also qualify for the job," says Kamat.

Other areas where aspirants could look to develop their skills are intelligent controls, sensor network, speech processing, monitor control and digital signal processing to name a few.

Nature of the job

"There are many entry-level roles in the field of instrumentation engineering. Prominent among them are sales engineer, application engineer, service engineer and marketing engineer," informs Jyotsna.

Jyotsna explains, "Sales engineers identify customer requirements and position the right product and solution from their portfolio to best align with